Lab 1: Requirements Engineering & UML Use-Case Modelling

Objective: From a short, instructor-provided scenario, elicit key functions and constraints; write clear, verifiable functional (FR) and nonfunctional (NFR) requirements; then translate them into a UML use-case diagram and one use-case flow.
Duration: 90 minutes

Software Requirements: Draw.io , LucidChart or any other online UML tool.

Deliverables

1. **Requirements Table** (Word/Excel):
    - Exactly **five FRs, out of which two are given** (FR-001) and **two NFRs** out of which **one** is already given. (NFR-001), each with:
        1. Req ID
        2. Type (Functional/Nonfunctional)
        3. Description ("The system shall…")
        4. Priority (High/Medium/Low)
        5. Acceptance Criteria (measurable pass/fail)
        6. Rationale (short justification)
2. **UML Use-Case Diagram** (PDF), showing all actors, use cases, and at least one «include» or «extend» relation.
3. **Use-Case Flow Document** (Word, one page):
    - For one key use case (e.g., "Place Order"), provide:

        - **Main Success Scenario** (step-by-step).
        - **Alternate Flow, at least one** (if payment is declined, for example).

Lab 1 Steps (90 min)

1. **Introduction & Scenario Review (5 min)** ○
   Students to work individually

The proposed system is an **online multi-vendor artisan marketplace** where independent craftspeople can create storefronts, manage their product catalogues, receive customer orders, and receive automated payouts.

The primary stakeholders identified in the problem statement are:

1. **Shopper** – browses products, adds products to the cart, places orders and makes payments.
2. **Artisan Vendor** – manages products/storefronts and receives earnings from sales.
3. **Payment Gateway** – an external system responsible for processing customer payments.

The problem statement specifically requires the marketplace to support **split payments**, where the amount due to multiple vendors is calculated from the customer's cart and the platform deducts a **5% platform fee** before allocating vendor earnings.

2. **Elicit & Draft Requirements (35 min)** ○
Brainstorm (10 min):

▪ Each student lists possible functions (e.g., “Select coffee type”) and constraints (e.g., “Order < 60 s”). ○ **Populate Requirements Table (25 min):** ▪ Use the template below.

- Fill in exactly **five FRs** and **two NFRs**.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-0 01	Functional	The system shall allow a customer to select a coffee type (Espresso, etc.)	High	When a customer taps “Espresso,” the screen highlights Espresso and displays “Size”	Core ordering functionality
FR-0 02	Functional	The system shall allow the customer to select a drink size.	High	When a size is selected, it is shown as chosen and the next step is prompted.	Required for price calculation and preparation
FR-0 03	Functional	The system shall allow artisan vendors to add, modify, and remove products from their storefront catalogue.	High	**Pass:** After a vendor adds, modifies, or removes a product, the corresponding catalogue state is displayed correctly. **Fail:** The catalogue does not reflect the requested operation.	Vendors need direct control over their product catalogues.
FR-0 04	Functional	The system shall allow shoppers to add products from multiple artisan vendors to a cart and submit the cart for checkout.	High	**Pass:** Products from at least two vendors can coexist in one cart and the system displays the correct combined	Supports the defining multi-vendor shopping experience

				cart value at checkout. **Fail:** Products cannot be combined or the cart total is incorrect.	
FR-0 05	Functional	The system shall allow artisan vendors to view customer orders containing their products and update the fulfilment status of those orders.	Medium	**Pass:** A vendor can view orders containing their products and change an order to an available status such as processing/shipped. **Fail:** The vendor cannot view or update the relevant order.	Allows vendors to manage orders received through the marketplace.
NFR001	Nonfunc.	The system shall complete any order (selection to payment) in under 60 s.	High	A timing log shows < 60 s from first tap to transaction approval	Ensures quick service during peak hours
NFR002	Nonfunc.	The system shall protect shopper payment and vendor payout information during transmission using secure communication protocols.	High	**Pass:** Security testing confirms that payment/payout information is transmitted only through encrypted connections. **Fail:** Sensitive financial information is transmitted through an unencrypted connection.	Protects sensitive financial information involved in customer payments and automated vendor payouts.

3. **Peer Critique & Revision (15 min)** ○ Swap requirement tables with another students.
    - Review each FR/NFR for:
        - **Clarity**: Is the “shall” statement unambiguous?
        - **Testability**: Does acceptance criterion yield a clear pass/fail?
        - **Relevance**: Does it align with the scenario?
    - Write short suggestions in a “Comments” column (e.g., “Need to specify which payment gateway or timeout”).
    - Swap back and **revise** your own table accordingly.

Requirement	Peer Critique	Revision
FR-001	Specify exactly how the financial values are validated.	Added reconciliation-based pass/fail acceptance criterion.
FR-002	"Search products" could be ambiguous.	Specified product name, category, and vendor as search dimensions.
FR-003	"Manage products" is broad.	Explicitly defined add, modify, and remove operations.
FR-004	Need to establish that products from different vendors can coexist.	Acceptance criterion explicitly tests a multi-vendor cart.
FR-005	"Manage orders" is too vague.	Defined viewing relevant orders and updating fulfilment status.
NFR-001	Must have a measurable performance threshold.	Retained the supplied `<500 ms` criterion.
NFR-002	"Secure" alone would not be testable.	Acceptance criterion specifies verification of encrypted transmission.

4. **Extract Actors & Use Cases (10 min)** ○ From the approved requirements, list:
    - **Actors (at least three)**

▪ Use Cases (at least five)

Actor 1 — Shopper

The person who searches for products, manages a cart, checks out and makes purchases.

Actor 2 — Artisan Vendor

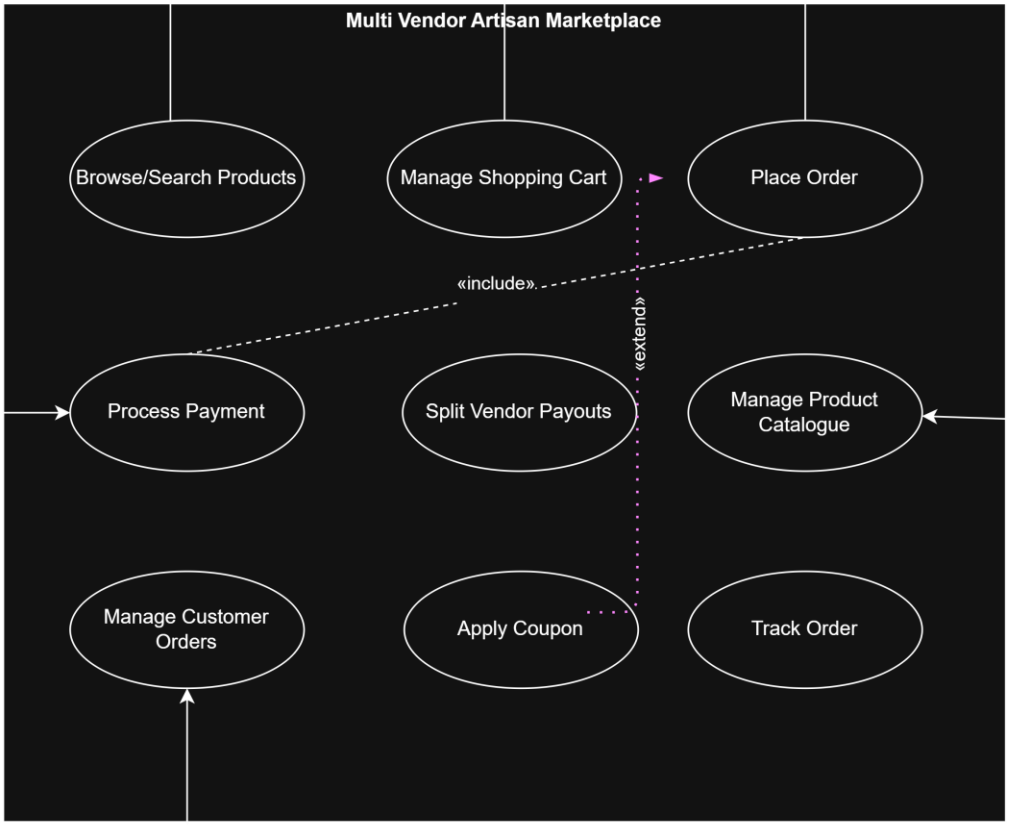
The independent craftsperson who manages products and fulfils customer orders.

Actor 3 — Payment Gateway

An external system that authorizes and processes the shopper's payment.

ID	Use Case	Main Actor(s)
UC-01	Browse/Search Products	Shopper
UC-02	Manage Shopping Cart	Shopper
UC-03	Place Order	Shopper
UC-04	Process Payment	Shopper, Payment Gateway
UC-05	Split Vendor Payouts	Payment Gateway / Marketplace System
UC-06	Manage Product Catalogue	Artisan Vendor
UC-07	Manage Customer Orders	Artisan Vendor
UC-08	Apply Discount/Coupon	Shopper
UC-09	Track Order	Shopper

5. **Draw at least one UML Use-Case Diagram (20 min)** ○ Open a UML tool (e.g., **draw.io** or **Lucidchart**).
    - **Actors** on left:
        - Draw stick figures: Actors ○ **Use Cases** (ovals) in center: label them UC-01…UC-05 with titles.
    - **Associations** ○ **Stereotypes**:
        - (At least one «extend»/<<include>> relationship is required: see step 6 below.)



6. **Write One Use-Case Flow (20 min)** ○ Pick a **UC:** In a separate page, draft:

    - **Main Success Scenario**
    - **Alternate Flow**

the best use case to select is:

UC-03 — Place Order

Preconditions

1. The Shopper has access to the marketplace.
2. At least one product is available in the shopping cart.
3. The selected products are available for purchase.
4. The payment gateway is accessible.

Postconditions

Success:

- The order is successfully created.
- Payment is processed.
- The 5% platform fee is deducted.
- Vendor earnings are allocated to the appropriate vendor accounts.
- The order is available for fulfilment.

Failure:

- No order is finalized.
- The shopper is informed that payment failed.

13. Main Success Scenario

UC-03: Place Order

Actor: Shopper
Supporting Actor: Payment Gateway

1. Shopper browses the marketplace and selects products.

2. Shopper adds one or more products to the shopping cart.
3. System displays the selected products and calculated cart total.
4. Shopper proceeds to checkout.
5. System identifies the artisan vendor associated with each cart item.
6. System calculates the total amount payable.
7. Shopper confirms the order and selects a payment method.
8. System sends the payment request to the Payment Gateway.
9. Payment Gateway authorizes the payment.
10. System calculates the **5% platform fee**.
11. System calculates the amount payable to each artisan vendor based on the products purchased from that vendor.
12. System records the order and associated vendor allocations.
13. System confirms that the split payout calculation reconciles with the customer's cart value.
14. System displays an order confirmation to the Shopper.
15. The use case ends successfully.

The critical step is the split-payout operation because it is the distinctive requirement supplied for this problem: the system must allocate item earnings to multiple independent vendor accounts after deducting the 5% platform fee.

14. Alternate Flow

AF-01 — Payment Declined

At **Step 9**:

1. Payment Gateway rejects the payment.
2. System displays a **"Payment Failed"** message.
3. System does not create a completed order.
4. Shopper is prompted to retry the payment using another available payment method.
5. If the retry succeeds, execution resumes at Step 10 of the Main Success Scenario.
6. If payment continues to fail, the system cancels the checkout attempt and retains the cart for the Shopper to try again later.
7. The use case terminates unsuccessfully.

Mapping Requirements → Use Cases

Requirement	Corresponding Use Case
FR-001 Split customer payments and allocate vendor earnings	**UC-05 Split Vendor Payouts**
FR-002 Browse/search products	**UC-01 Browse/Search Products**
FR-003 Vendor product management	**UC-06 Manage Product Catalogue**
FR-004 Multi-vendor cart and checkout	**UC-02 Manage Shopping Cart / UC-03 Place Order**
FR-005 Vendor order management	**UC-07 Manage Customer Orders**
NFR-001 <500 ms catalogue/image delivery	**UC-01 Browse/Search Products**
NFR-002 Secure payment/payout transmission	**UC-04 Process Payment / UC-05 Split Vendor Payouts**